A messy Downloads or Desktop folder is something almost everyone deals with. We can fix that completely using Python.
The script below uses Python's built-in os and shutil modules. It looks at the file extensions (like .pdf or .png), creates target folders if they don't already exist, and moves the files safely.

The Python File Organizer

```
import os
import shutil

# Define the path of the folder you want to clean up.
# Use '.' for the current folder where the script runs, or provide an
absolute path.
# Example: TRACK_DIRECTORY = r"C:\Users\YourName\Downloads"
TRACK_DIRECTORY = "."

# Map file extensions to their corresponding target folder names
FILE_EXTENSIONS_MAP = {
    # Documents
    '.pdf': 'Documents',
    '.docx': 'Documents',
    '.doc': 'Documents',
    '.txt': 'Documents',
    '.xlsx': 'Documents',
    '.pptx': 'Documents',

    # Images
    '.jpg': 'Images',
    '.jpeg': 'Images',
    '.png': 'Images',
    '.gif': 'Images',
    '.svg': 'Images',

    # Audio/Video
    '.mp4': 'Media',
    '.mkv': 'Media',
    '.mp3': 'Media',
    '.wav': 'Media',

    # Archives/Zip files
    '.zip': 'Archives',
    '.tar': 'Archives',
    '.gz': 'Archives',
    '.rar': 'Archives',

    # Executables/Installers
    '.exe': 'Installers',
    '.dmg': 'Installers',
```

```
    '.msi': 'Installers'
}

def organize_folder(target_dir):
    """
    Scans the target directory and moves files into organized folders
    based on their file extensions.
    """
    # Verify if the provided path actually exists
    if not os.path.exists(target_dir):
        print(f"Error: The directory '{target_dir}' does not exist.")
        return

    print(f"Scanning and organizing: {os.path.abspath(target_dir)}\n")
    moved_count = 0

    # Iterate through all items in the specified directory
    for filename in os.listdir(target_dir):
        file_path = os.path.join(target_dir, filename)

        # Skip directories; we only want to organize individual files
        if os.path.isdir(file_path):
            continue

        # Skip the script itself if it's running inside the target
directory
        if filename == os.path.basename(__file__):
            continue

        # Extract the file extension and convert it to lowercase
        _, file_extension = os.path.splitext(filename)
        file_extension = file_extension.lower()

        # Check if the extension is mapped to a folder destination
        if file_extension in FILE_EXTENSIONS_MAP:
            folder_name = FILE_EXTENSIONS_MAP[file_extension]
            folder_path = os.path.join(target_dir, folder_name)

            # Create the folder dynamically if it doesn't exist yet
            if not os.path.exists(folder_path):
                os.makedirs(folder_path)
                print(f"Created folder: {folder_name}")

            # Define the final destination path for the file
            destination_path = os.path.join(folder_path, filename)

            try:
                # Move the file safely
```

```
                shutil.move(file_path, destination_path)
                print(f"Moved: {filename} -> {folder_name}/")
                moved_count += 1
            except Exception as e:
                print(f"Failed to move {filename}. Error: {e}")

    print(f"\nCleanup complete! Total files moved: {moved_count}")

if __name__ == "__main__":
    organize_folder(TRACK_DIRECTORY)
```

How to use this script

1. **Save the Code**
   File setup
   Copy the code block above and paste it into a file named file_organizer.py.
2. **Set the Target Folder**
   Configuration
   Change TRACK_DIRECTORY = "." to the specific path you want to clean up. For example, if you're on Windows, it might look like r"C:\Users\YourUsername\Downloads". The r before the quote handles backslashes correctly.
3. **Run the Script**
   Execution
   Open your terminal or command prompt, navigate to where you saved the file, and run:

Safety Check: The script includes a safeguard that ignores directories entirely, meaning it won't mess with folders you have already organized manually. It also automatically skips moving itself!